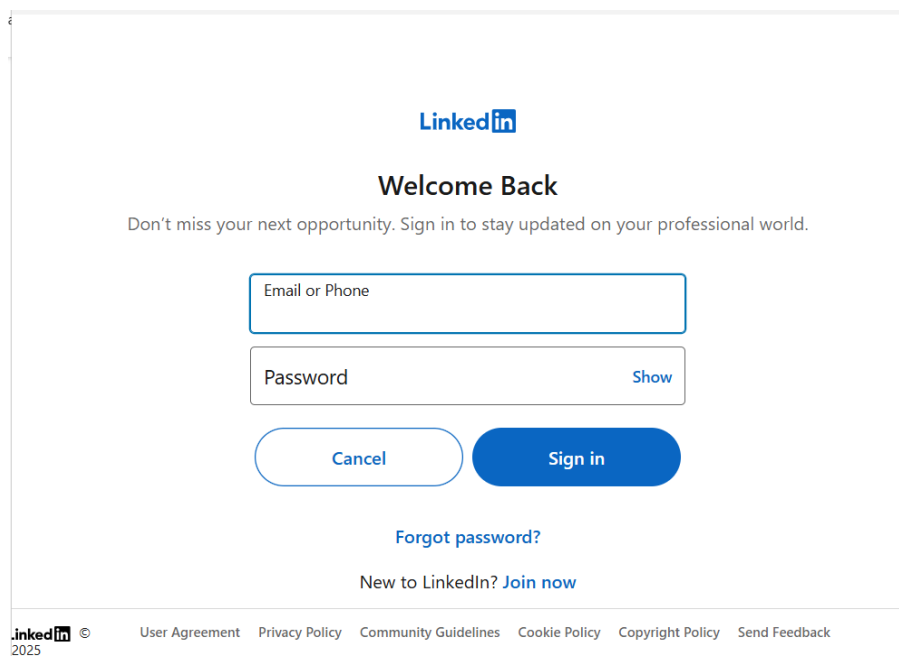
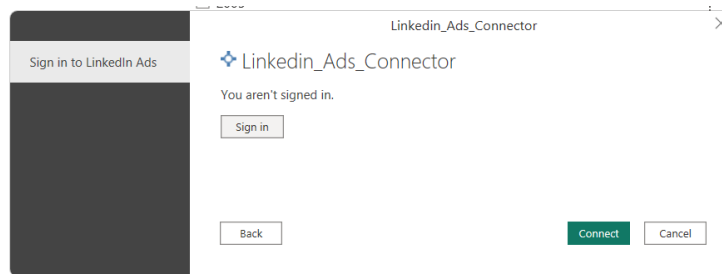


1. Power BI Desktop / Power BI Service UI



2. OAuth Start Phase (StartLogin Handler)



- Power BI calls StartLogin()
- Connector constructs OAuth Authorization URL using Config values
- Power BI opens LinkedIn's OAuth Authorization Page



3. User Authentication on LinkedIn

- User logs in and approves the scopes
- LinkedIn redirects to RedirectUri with ?code=<authorization_code>



4. OAuth Token Exchange Phase (FinishLogin Handler)

- Power BI passes redirect URL to FinishLogin()
- Connector exchanges authorization_code for:
 - access_token
 - refresh_token
 - expires_in
 - token_type
 - scope



5. Secure Credential Storage (Power BI Credential Vault)

- Power BI securely stores the returned OAuth tokens in its credential store:

Power BI Desktop:

- Stored in Windows Credential Manager
- Encrypted using DPAPI, tied to Windows user profile

Power BI Service:

- Stored in the Power BI Cloud Credential Store
- Backed by Azure Key Vault with managed encryption



6. Connector Execution (Linkedin_Ads_Connector.Contents)

- User selects a data object or performs refresh
- Power BI loads the connector function



7. Credential Retrieval (Extension.CurrentCredential)

- Power BI retrieves the stored token from its credential vault

- Connector receives:

```
[  
  access_token = "..."  
  refresh_token = "..."  
  token_type = "Bearer"  
  expires_in = 3600  
  ...  
]
```



8. API Request Phase (GetLinkedInAnalytics)

- Connector sends request to LinkedIn REST Ad Analytics API
- Authorization header includes:

Authorization: Bearer <access_token>



9. LinkedIn API Response Handling

A) If **HTTP 200 OK**:

- Response is parsed
- Data transformed into a Power BI Table

B) If **HTTP 401 Unauthorized (Access Token Expired)**:

- Power BI intercepts the 401 error
- Power BI triggers OAuth Refresh Flow automatically:



10. OAuth Refresh Phase (RefreshToken Handler)

- Power BI calls RefreshToken()
- Connector sends:

grant_type=refresh_token

refresh_token=<stored_refresh_token>

client_id / client_secret

- LinkedIn returns:
 - new_access_token
 - new_refresh_token (if rotated)
 - new expires_in metadata



11. Token Replacement in Credential Store

- Power BI securely replaces the old tokens with the new ones:

Desktop → Credential Manager

Service → Azure Key Vault-backed Credential Store



12. Automatic Retry of Original Request

- Power BI retries the original LinkedIn API request
- New access token is used



13. Successful LinkedIn API Response

- JSON data is parsed
- Data transformed into columns
- Result returned to Power BI



14. Final Output

- Power BI displays the final table to the user